

Research Summary

Title: Finding Efficiency in Approach Selection in Software Development and Project Management in Bangladesh IT Industry

Objective:

This study investigates the effectiveness of different software development methodologies (Agile, Waterfall, etc.) in Bangladesh's IT sector, analyzing how a company's maturity level influences its choice of methodology and project success.

Key Findings:

1. Agile is the Preferred Choice:

- **Startups (<2 years old):** 95% use Agile (primarily Scrum) due to its adaptability, faster delivery cycles, and ability to accommodate changing requirements. These companies report high satisfaction.
- **Growing Companies (2–5 years old):** Results are mixed—while some continue benefiting from Agile, others struggle with scaling or unclear project scopes.
- **Established Companies (5–8 years old):** ~90% rely on Agile, leveraging its iterative approach for stable project management and quality assurance.
- **Mature Companies (8+ years old):** Use a combination of Agile and traditional methods (e.g., Waterfall). About 40% are open to transitioning fully to Agile.

2. Why Companies Prefer Agile:

- **Flexibility:** Enables quick adaptation to market and client demands.
- **Cost Efficiency:** Reduces financial risks through incremental development.
- **Quality Improvement:** Continuous testing and feedback ensure higher product standards.
- **Client Satisfaction:** Regular stakeholder involvement aligns development with expectations.

3. Challenges Identified:

- **Mid-sized firms (2–5 years):** Face difficulties in scaling Agile for larger teams or complex projects.
- **Older firms (8+ years):** Often modify Agile or blend it with traditional methods to fit long-term projects.

Recommendations:

- **New Companies:** Start with Agile (Scrum or Kanban) for flexibility and rapid delivery.

- **Growing Companies:** Invest in Agile training and consider hybrid models for smoother scaling.
- **Mature Companies:** Use Agile for innovation-driven projects and traditional methods for stable, long-term developments.

Future Work:

The study suggests expanding research to include more companies across different regions, refining survey questions for deeper insights, and developing a decision-making framework to help firms choose the best methodology based on their specific needs.

Conclusion:

Agile is the most effective methodology for most IT companies in Bangladesh, particularly startups and mid-sized firms. However, mature companies benefit from a tailored approach, combining Agile with traditional methods when necessary. By selecting the right methodology, firms can enhance efficiency, reduce costs, and deliver higher-quality software products.